IAN ZEPP

Principal / Staff Engineer · Software Architect · AI Agent Infrastructure and Developer Tools
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U.S. citizen, no sponsorship required · U.S. Marine Corps veteran · Previously held a U.S. Secret clearance

SUMMARY

Principal engineer and software architect who builds AI agent infrastructure, compilers, and multi-tenant platforms, and who has led engineering organizations. Since August 2025, building full time in Rust, Go, and TypeScript: a statically typed language and compiler for coding agents (Faber), distributed dispatch for agentic engineering jobs across a multi-machine cluster (Vivi), an MCP governance gateway and security firewall (Cephalopodic), and behavioral evals of current models. Works at depth with coding agents as a full-time daily practice (Claude Code, Codex, self-built harnesses). Decades of Linux; RHCE. Earlier: led platform R&D at Motivis Learning as VP; VP of Software Engineering at roundCorner, promoted from Chief Scientist; solo founder of Minted.IO, a multi-tenant platform Motivis acquired; and forward-deployed Salesforce technical solution architect. Left the industry in 2021 and returned to full-time engineering in August 2025.

TECHNICAL SKILLS

AI and agent systems: LLM orchestration, multi-agent systems, agent harnesses and tool calling, Model Context Protocol (MCP), multi-provider model routing, LLM proxy and credential pools, evaluation harness design, behavioral and performance evals, retrieval-augmented generation, durable agent memory across context compaction; daily tools: Claude Code, Codex, Cursor, custom agent harnesses; operator of a fleet of specialized AI agents (Grok Bot assistant and engineering bots, scheduled routines, cloud agents)

Open-weight and local models: local GGUF inference on llama.cpp (llama-bench and llama-server runtime tuning), self-hosted model upstreams behind an OpenAI-compatible proxy, open-weight models as eval subjects and judges

Security and governance: MCP governance gateway and policy enforcement, multi-tenant isolation, record-level ACLs, dead-credential detection, checksum-verified installers

Languages: Rust, Go, TypeScript, JavaScript, Node.js, Bun, Python, SQL, Java, PHP, Swift, Bash; earlier career: C/C++ and Qt (KDE contributor, 1999)

Indirect (via Faber code generation): working knowledge of Rust, TypeScript, Go, Swift, Python, Haskell, C, Racket, LLVM IR, Cranelift IR, WebAssembly, WGSL, and GPU kernels for Metal and CUDA (via LLVM)

Front end: Rust/WebAssembly with Leptos (SSR and hydration), React, Angular, TypeScript, HTML and CSS

Compilers and developer tools: language design, type systems, HIR/MIR lowering, code generation, cross-compilation, GPU kernel code generation (Metal, CUDA), differential testing, tree-sitter grammars, CLI tooling, Git

Platforms, infrastructure, and operations: Linux (decades; Red Hat Certified Engineer, RHCE), backend services, full-stack web applications, distributed systems, multi-tenant architecture, HTTP and JSON-RPC APIs, PostgreSQL, MySQL, SQLite, Docker, CI/CD and automated build-and-test loops, monitoring and observability, AWS, Google Cloud (including Spanner), Fly.io, RunPod (GPU), Salesforce platform (Apex, Visualforce, managed packages)

Leadership and delivery: technical roadmap, platform and product architecture, hiring, engineering organization design, forward-deployed enterprise delivery, live production incident response

EXPERIENCE

Independent AI Systems Engineer, Self-employed — 08/2025 – Present · Asheville, NC · Remote

- **Agent operations:** Runs engineering through coding agents. Planning, implementation, and audit are separate roles, and each audit is run independently by two models from different families plus a synthesizer.
- **Faber and Radix (faberlang.dev):** Designed Faber, a statically typed language for coding agents and human authors, and built Radix, its Rust compiler: parser, type system, HIR and MIR lowering, and code generation from Rust, TypeScript, Go, and Swift to LLVM IR, Cranelift, WebAssembly, and WGSL, with a public documentation site. GPU kernel generation: @kernel functions lower to Metal and CUDA (via LLVM) and launch on real GPUs, with fail-closed device execution; the autograd library uses Faber-written RMSNorm, GEMM, RoPE, softmax, and SwiGLU kernels. In progress, not released: multi-GPU/cluster kernel execution, partitioning work by memory chunk or thread lane. Replaced the TypeScript compiler with Rust in one release, deleting 247,791 lines with no loss of test coverage.
- **Vivi (github.com/ianzepp/vivi, open source):** Built distributed dispatch for agentic engineering jobs across a multi-machine cluster. The pump, an unattended state-machine dispatcher, routes planner, implementation, merge, gate, and audit jobs to role-based worker pools on local Git worktrees, Docker, SSH hosts, Fly.io, and RunPod GPU machines, with fallback routing chains and per-role model bands, orchestrated from a Linux workstation hub on a home network that runs the pump and proxy. vivi proxy is a loopback LLM front door that routes OpenAI- and Anthropic-style APIs to multiple providers and local llama.cpp servers, with credential pools, token refresh, dead-credential detection, and supervised child processes. Underneath, a local-first mailspace keeps tasks, memos, and work graphs durable across restarts, context compaction, and model hand-offs.
- **Evals (llm-pressure-trials, Faber trials, llama-gguf-tune):** Behavioral red-team evals of model responses to framing shifts, invented authority, and coercion on current models, with pre-registered predictions, a 2x2 grader audit, and a grading stack validated at 96%

against hand-labeled gold. Faber learnability trials: n-shot matrices and executable grading. Local GGUF inference autotuning on llama.cpp (llama-bench matrices, llama-server evals).

- **Cephalopodic (Cephi):** Co-founded with two Gauntlet AI peers as the capstone: a Go control plane and security gateway for MCP (registration, catalogs, user-scoped taps, policy enforcement, JSON-RPC inspection), a defensive firewall between agents and tool servers. Wound down within about two months once managed MCP shipped natively from the model vendors.
- **Monk platform and Monk OS:** monk-api, the third clean-room build of the schema-as-data, multi-tenant platform behind Minted.IO and Motivis: Hono, TypeScript, PostgreSQL, tenant isolation, observer rings, generated REST APIs, plus a Rust CLI, Go FUSE filesystem, FTP and IRC bridges, an MCP server, and the monk-bot agent daemon. Monk OS: a Plan 9-inspired OS kernel in TypeScript on Bun where the AI agent is PID 1: a VFS where every file is a queryable database record, syscalls, Worker-isolated processes, a HAL, SQLite or PostgreSQL storage, and an SDK.
- **Also:** Field Board, a real-time collaborative whiteboard in Rust end to end (Axum, PostgreSQL, Leptos/WebAssembly, protobuf), with chat and an AI agent that edits the board through tool calls; Minted Geek Swarm, a Rust multi-agent runtime where agent fleets provision their own mail, CRM, storage, payments, and MCP tool servers; LegacyLens, retrieval-augmented analysis of a COBOL codebase with hybrid vector retrieval.

Member of Technical Staff, Kraken 01/2021 – 03/2021 · Contract

- Money-movement systems for a global crypto exchange: SWIFT bank transfers and wire-transfer verification.

Vice President, Platform Research and Development, Motivis Learning 02/2017 – 03/2020 · Salem, NH

- Led platform R&D across 10 to 20 engineers in a company backed by Southern New Hampshire University, built on the Minted.IO platform after its acquisition. Owned the technical roadmap and scaled the engineering organization.
- Extended the platform package model to cross-tenant namespace subscriptions: schools held private tenant data alongside read-only access to centrally managed curriculum and competency data, exposed through visibility rules. Delivered deployment portability across AWS, Google Cloud (Spanner), and local environments on TypeScript, Node.js, and PostgreSQL. The company closed in Q1 2020.

Technical Solution Architect, Salesforce 02/2016 – 02/2017

- Forward-deployed technical solution architect embedded on site with enterprise clients across two six-month assignments. CarMax: delivered a portal for on-site sales agents to track customer purchases, integrating multiple Salesforce products. Home Depot: implemented Field Service in its first release and integrated it with nightly batch processing.

Founder & Principal Engineer, Minted.IO 07/2015 – 12/2015 · Nashua, NH

- Designed and built solo, in six months, a metadata-driven, multi-tenant application platform; Motivis Learning bought the software and API assets for $750,000 in February 2017 as the foundation of its learning management system. Schema stored as data, so changes propagated immediately across every data API route; single-database PostgreSQL multi-tenancy with self-service provisioning, record-level ACLs (read, edit, group-based deny), and a recursive query DSL.

Vice President of Software Engineering (promoted from Chief Scientist), roundCorner 2011 – 2015 · New York, NY

- Senior technical authority for a Salesforce-native enterprise CRM with no separate CTO. Led a direct team of engineers and project managers alongside an offshore delivery organization, shipping fundraising, grant-making, and alumni systems for St. Jude Children's Research Hospital, Georgetown, Duke, Villanova, and Girl Scouts of the USA.
- Owned architecture for a fundraising and constituent management product line adopted by more than 200 nonprofits, foundations, and universities, later acquired by Salesforce.org. Designed the migration data model off Blackbaud Raiser's Edge and Team Approach, and built versioned namespaced managed packages where released metadata can never be removed.
- Diagnosed a live production failure during a St. Jude Radiothon (concurrent donation entry triggered cascading child-record generation that saturated write throughput), traced it in platform debug logs during the event, and applied a reversible mitigation.

EARLIER EXPERIENCE

Founder & CTO, Minted Geek Hungary KFT, Budapest (2013 – 2014). Salesforce consulting company co-founded with a local partner.

Founder & Principal Consultant, M43 Instead LLC (2010 – 2011). Salesforce consulting (Bluewolf, MaidPro, roundCorner, others). For MaidPro, replaced spreadsheet estimating across its franchise offices with a pricing, scheduling, and dispatch portal.

Technical Architect, Red Hat, Raleigh-Durham, NC (2009 – 2010). Multilingual APAC Partner Center (Korea, Japan, China).

Software Engineer, Beatport, Denver, CO (2008 – 2009). PHP, enterprise Java, Mule, Spring, Hibernate, Squid, Akamai, and MySQL.

Real Estate Sales, Bosshardt Realty Services, LLC, Gainesville, FL (2002 – 2007)

Open-source contributor, KDE (1999 – 2000). Wrote KStringHandler, accepted into kdelibs/kdecore; KMap, a C++/Qt Nmap front end credited in the Nmap 2.50 release notes; a Python Gnutella P2P library.

Infantry Team Leader (E-4), United States Marine Corps (1996 – 2000). Honorable service; Corporal (E-4), 0351 Assaultman.

EDUCATION

Bachelor of Business Administration, UMass Amherst. **Languages:** English; conversational French and Spanish

AI Engineering Fellowship, Cohort 4, Gauntlet AI, Austin, TX (2026): ten weeks, continuous elimination.

Certifications (2007 to 2009): RHCE, SCJP, CCNA, CCENT, CompTIA Security+ and A+, Zend Certified Engineer, MySQL Certified Developer